and twenty million light-years away. That's a completely untestable prediction, of course, but we maintain a very high credence that it's true. There's no reason for gravity to behave in that way, and imagining that it did leaves us with a much uglier theory than the one we have. The additional worlds in Everettian quantum mechanics have exactly this character: they are inescapable predictions of a simple theoretical formalism. We should accept them unless we have a specific reason not to.

"And besides," Alice rushed on, "the other worlds could be detected in principle, if we got incredibly lucky. They haven't gone away, they're still there in the wave function. Decoherence makes it fantastically unlikely for one world to interfere with another, but not metaphysically impossible. I wouldn't suggest applying for grant money to do such an experiment, though; it would be like mixing cream into coffee and waiting around for them to spontaneously unmix themselves."

"Don't worry, I wasn't planning on it. I just don't think Karl Popper would be very happy with your approach to the philosophy of science."

"I've got you there, Dad," said Alice. "Popper himself was a harsh critic of the Copenhagen interpretation, which he called a 'mistaken and even a vicious doctrine.' In contrast, he had good things to say about Many-Worlds, which he accurately described as 'a completely objective discussion of quantum mechanics.'"

"Seriously? Popper was an Everettian?"

"Well, no," Alice admitted. "He ultimately parted ways with Everett because he couldn't understand why the wave function would branch but branches wouldn't later fuse back together. I mean, that's a good question, but it's one we can answer."

"I'm sure you can. Where did he come down on the foundations of quantum mechanics?"

"He developed his own formulation of quantum mechanics, but it never really caught on."

"Ha! Philosophers."

"Yeah. We're better at telling you why your theory is wrong than at proposing better ones."